

PRANAV SHIROLE

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B. Tech in Electronics and Telecommunication

Nov 2022 – July 2026

Sardar Patel Institute of Technology

CGPA: 8.10

Minor in Computer Engineering

Aug 2023 – Dec 2025

Sardar Patel Institute of Technology

Technical Skills

Languages: Python, C++, Javascript

Web Development: Django, FastAPI, REST API, React, Websockets, Tailwind

Databases: MySQL, PostgreSQL, Redis

Development Tools: Github, Postman, Selenium

Projects

Watch Party | *Django, REST API, React*

May 2025

- Created a platform for synchronous video playback and real time chat.
- Created a **REST API** using **Django Rest Framework** and used **Postman** to test the endpoints.
- Used **Axios Interceptors** and **JSON Web Tokens** for stateless authentication and authorisation.
- Used **RabbitMQ** and **Celery** to asynchronously process uploaded videos.
- Remuxed uploads into multiple qualities using FFmpeg to ensure smooth playback and reduced buffering.
- Implemented synchronised playback and real time chat using **WebSocket connections**.
- Designed a responsive and accessible user experience implementing login and upload forms, video player and scrollable chat.

Samarth AI (Smart India Hackathon 24) | *Python, ChromaDB, FastAPI, React*

Dec 2024

- Developed a ChatBot to aid employees search and understand technical documents and HR policies using NLP, thereby boosting their productivity.
- Implemented **RAG model** to retrieve relevant information from lengthy documentation and policies and generate response using Llama3 8B.
- Split the information into chunks and added **Contextual Chunk Headers** to improve accuracy.
- Used **Key Cycling** and **Multi-Threading** to reduce initial processing time of documents.
- Collaborated with team to ensure seamless integration of all features.

Fashion Recommendation System | *Tensorflow, Pandas, Django, React*

April 2024

- Created a Clothing E-Commerce website with matching clothes recommendation feature.
- Analysed the Kaggle H&M User Transactions dataset using **Matplotlib**, identifying outliers.
- Used **Pandas** to clean the dataset and reduce size by uniform sampling.
- Implemented **Collaborative Filtering** using **Tensorflow** to generate clothing recommendations.

8-bit CPU | *Computer Architecture, Microprocessors*

May 2024

- Designed an 8-bit CISC Processor in Logisim Circuit Simulator.
- Created an Instruction Set of 32 instructions to support simple programming.
- Components include 8-bit ALU, 4 GPRs, 256B RAM, ROM, Control Unit, Instruction Decoder and Program Loader.

Achievements

Smart India Hackathon 2024: First Place among 150+ teams from all over India.

Dec 2024

CodeUnicode: Finalist in Competitive Programming Competition held by top colleges in Mumbai.

April 2024

E-Summit BizQuest: Second Place in Business Case Study to revive failing Airline company.

July 2024

Code Red: Second Place in Tech based Treasure Hunt by GDSC.

July 2024

Positions of Responsibility

Sports Committee, SPIT

Sept. 2023 – Present

Executive Head

SPIT, Mumbai

- Conducted large scale sports events, overseeing event planning and team management.
- Organised SPIT Half Marathon in Juhu with 900+ runners.
- Coordinated Inter-College Sports Fest with 8 sports and 1200+ sportsmen over 11 days.
- Hosted International Chess Tournament attracting 400+ players.

Training and Placement Office

Sept. 2023 – Oct. 2024

Training and Placement Coordinator

SPIT, Mumbai

- Coordinated with placements drives and worked with companies to help with placement process.
- Organised and maintained student and placement data spanning over 5 years.